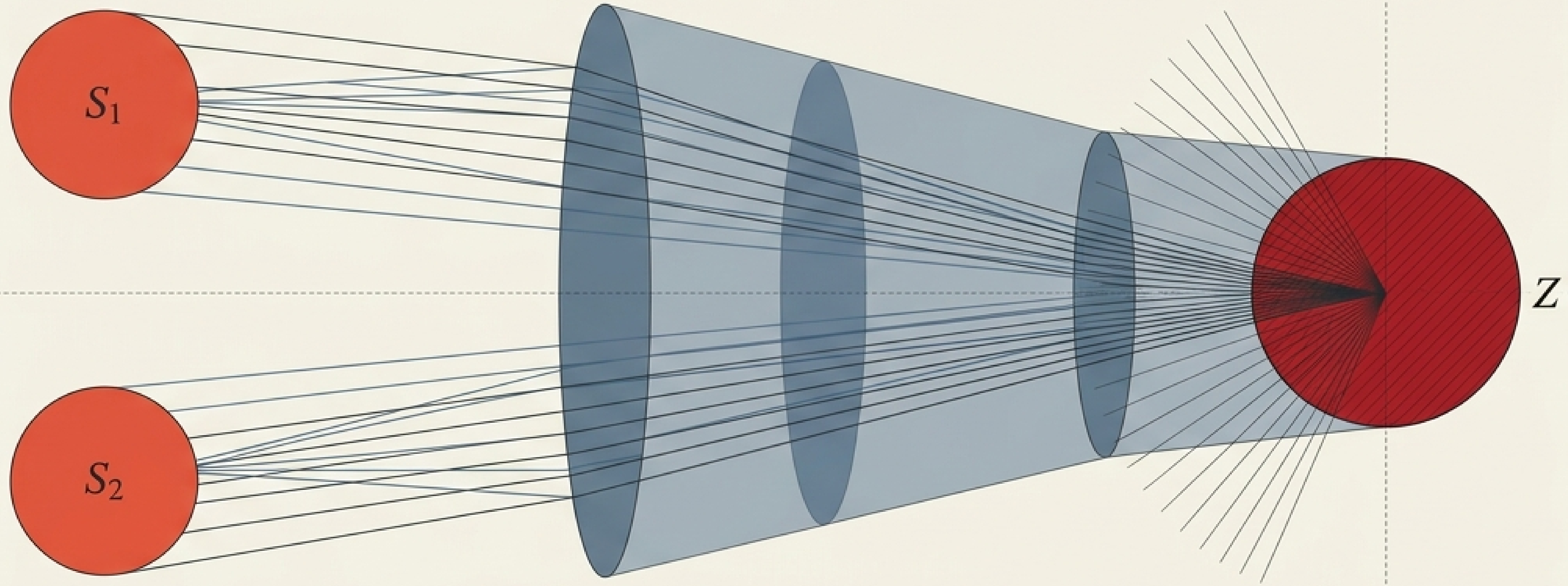
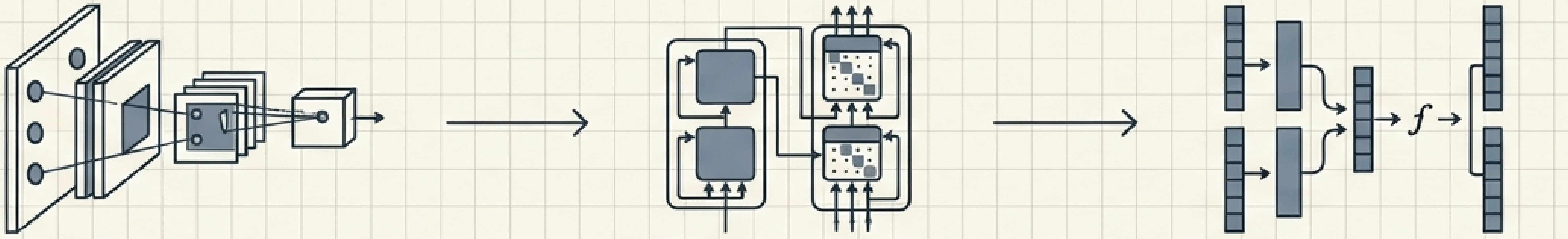
Against Latent Fundamentalism

The Missing Criterion in Representation Learning for Planning

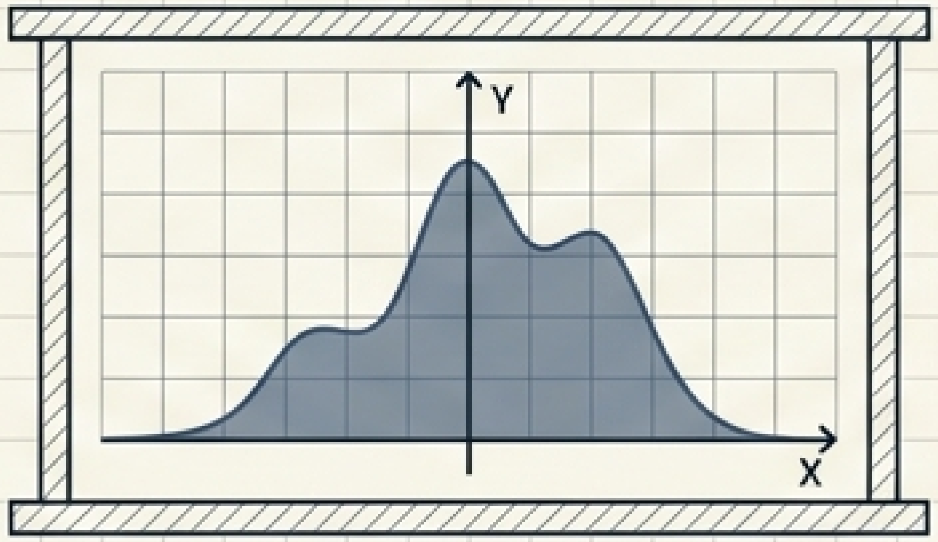


Why predictive and compressive world models structurally fail at planning—and how Admissibility Distortion provides a mathematical path forward.

The Shared Blind Spot in AI Planning



CNNs — Transformers — JEPAs (Joint Embedding Predictive Architectures)

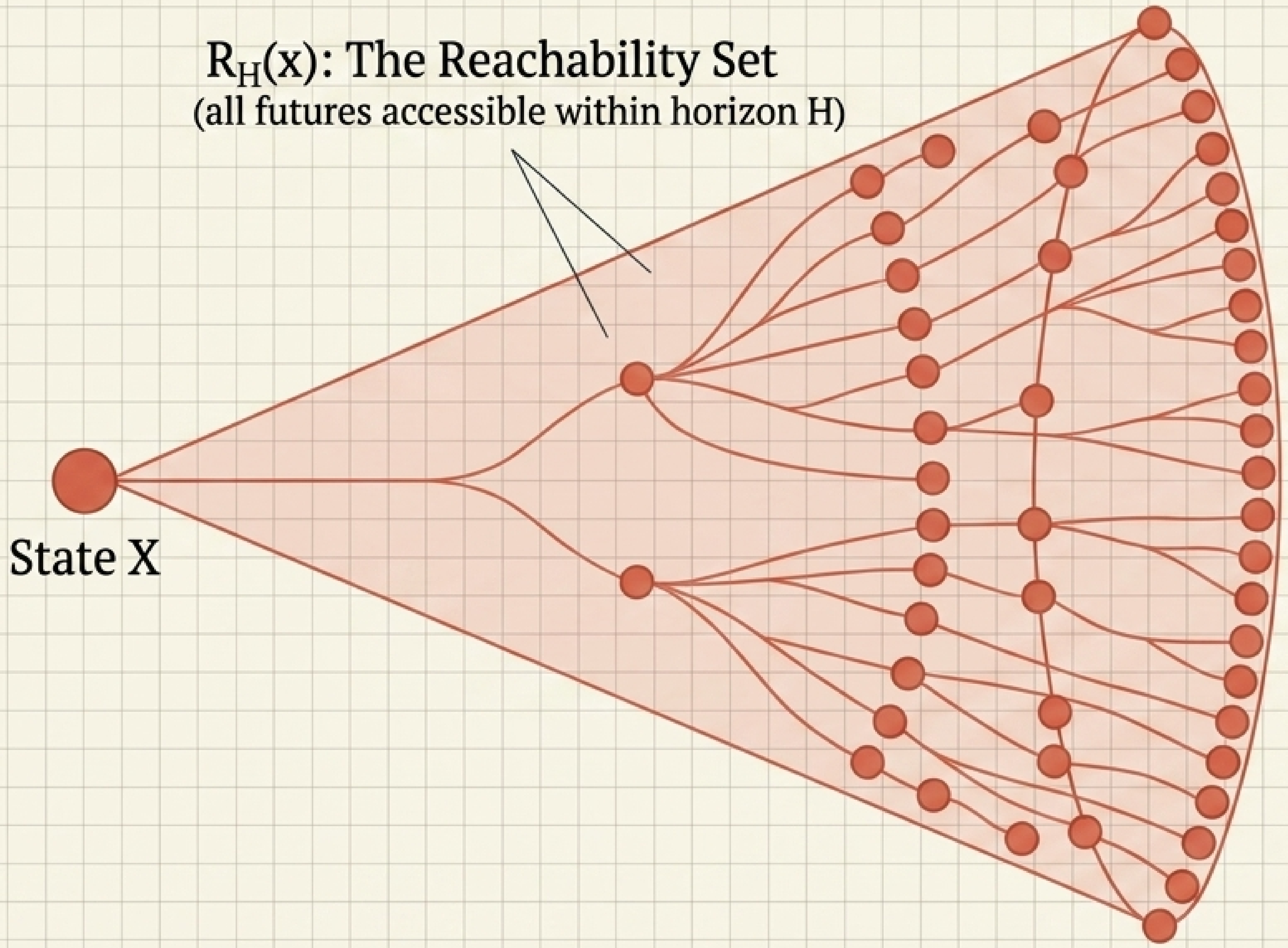


THE OPTIMIZATION ERROR

They all optimize for properties of the observation distribution (prediction or compression), not the preservation of reachability structure.

Moving prediction from pixel space to latent space changes the domain, but it does not fix the *fundamental category error*.

The First Principle of Planning: Reachability



Proposition 1

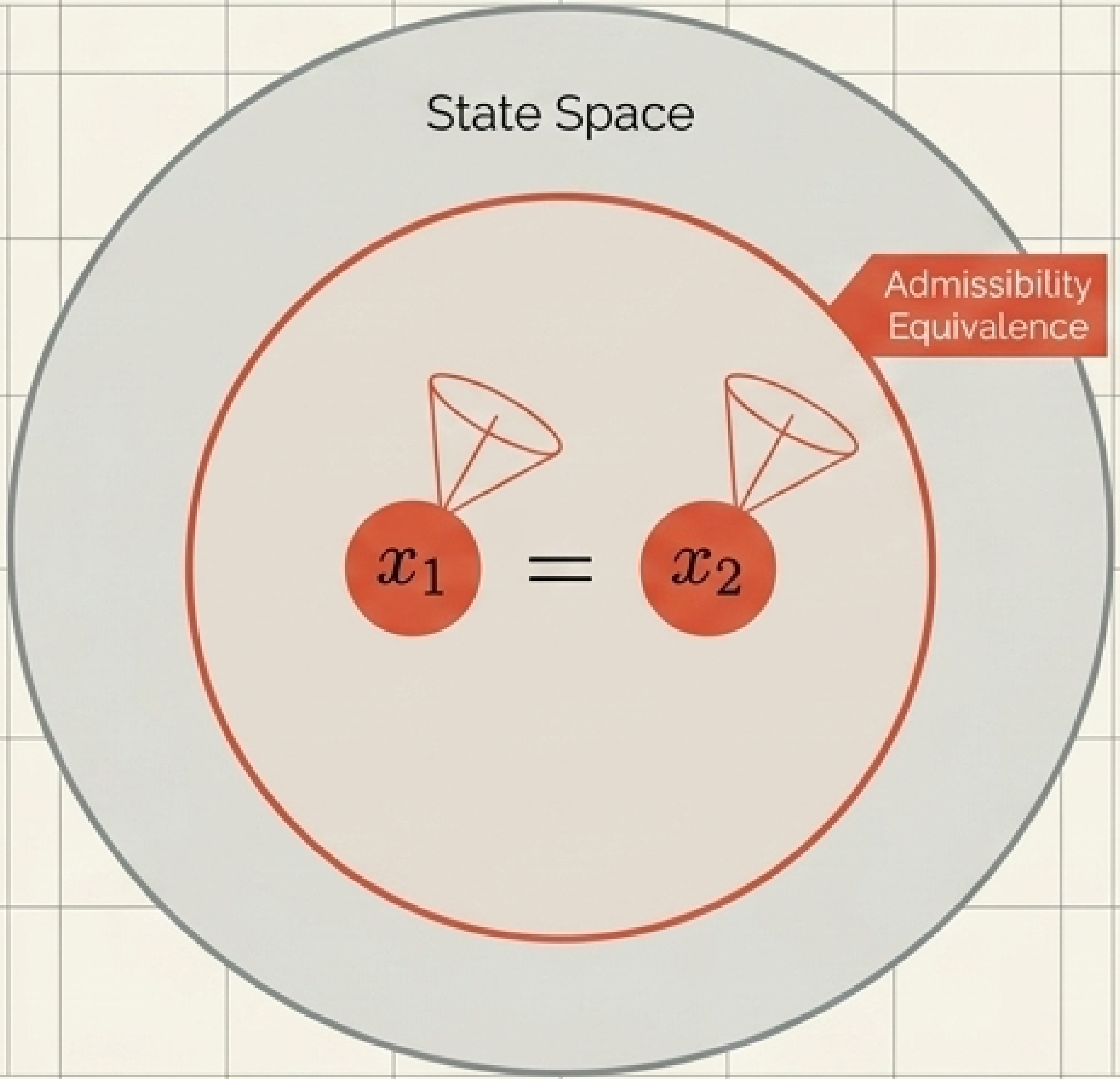
If two states have different reachable futures, a representation must distinguish them.

The Implication

Before an agent can optimize a reward, it must know what futures are accessible.

Collapsing states with different reachabilities guarantees planning errors that no optimizer can fix later.

Defining the Canonical Target: Admissibility Equivalence



The Definition

Two states are admissibility-equivalent IF AND ONLY IF they share the exact same reachable futures.

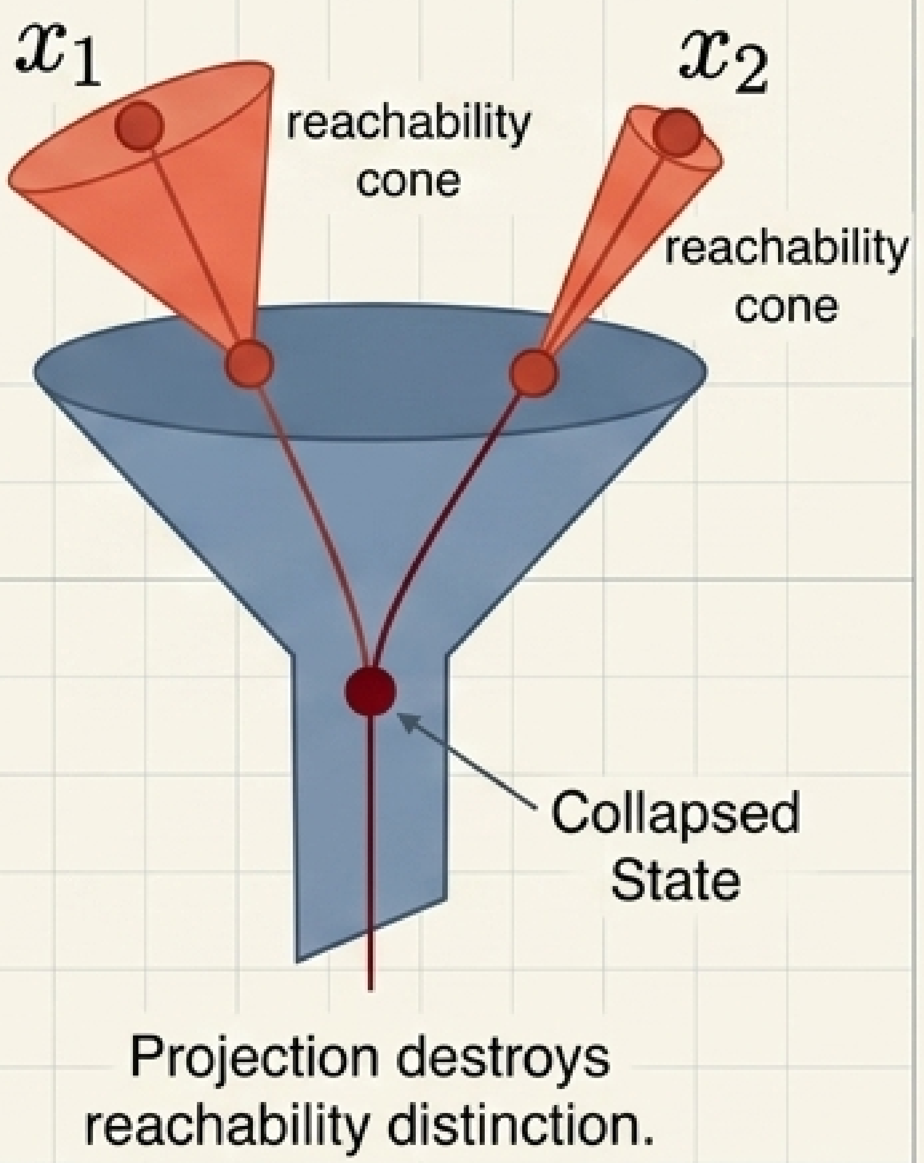
The Rule

This is the coarsest safe relation. A representation may safely collapse any pair in this set. It may never safely collapse any pair outside it. It is the minimal structural property presupposed by every planning problem.

Admissibility Distortion

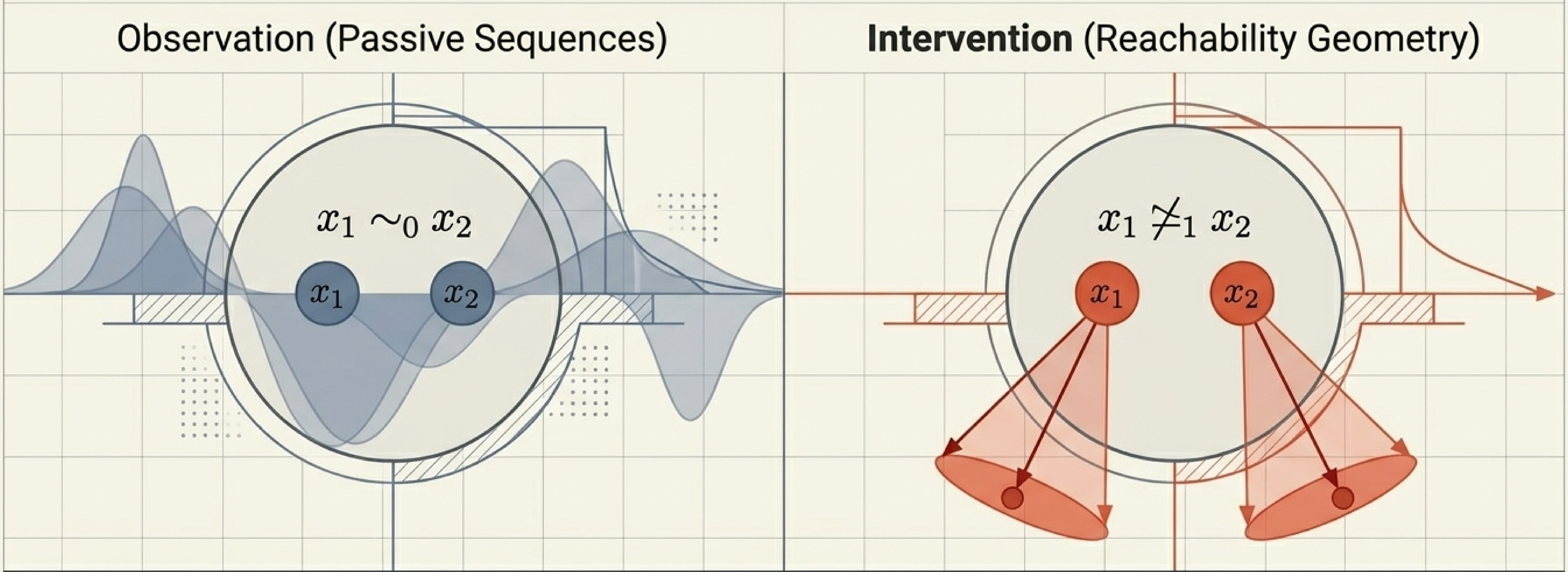
$$D_A(\phi) = \mathbb{E}_{(x_1,x_2)\in S^2}[\mathbb{I}[\phi(x_1) = \phi(x_2)] \cdot d_R(R_H(x_1), R_H(x_2))]$$

1. Expectation over state pairs.
2. The Penalty: Activated only when a learned projection collapses two states.
3. The Multiplier: The geometric distance between their actual reachable futures.



$D_A > 0$ whenever a learned projection destroys necessary reachability distinctions.
This is the exact quantity that world models fail to preserve.

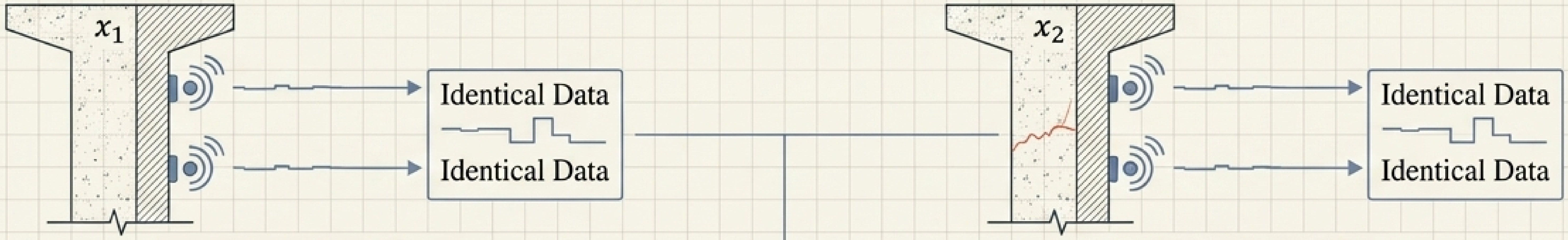
The Category Mismatch: Observation vs. Intervention



The Central Theorem: Observational equivalence does not imply interventional equivalence.
The Conclusion: They are generated by operations of different mathematical types. You cannot fix this category mismatch by simply collecting more data or moving prediction to a latent space.

The Separation Theorem in the Physical World

Phase 1: Passive Observation



Sensors read identical data.
$P(O|x_1) = P(O|x_2)$.

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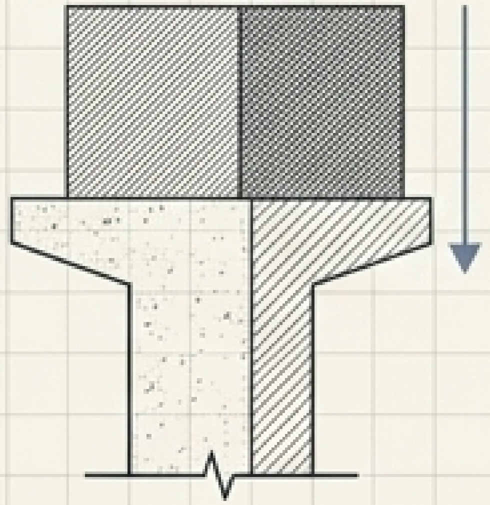
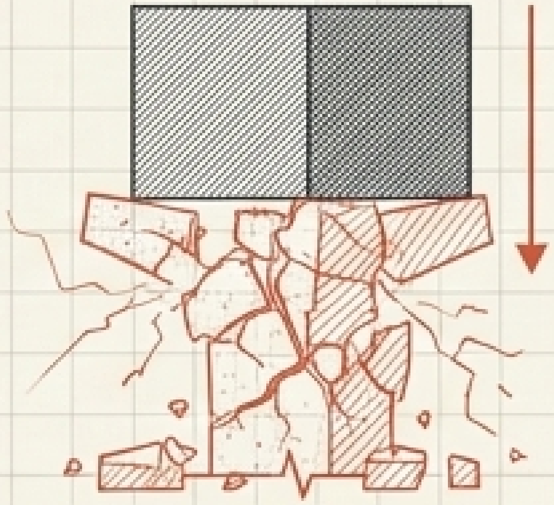
Phase 2: Interventional Planning



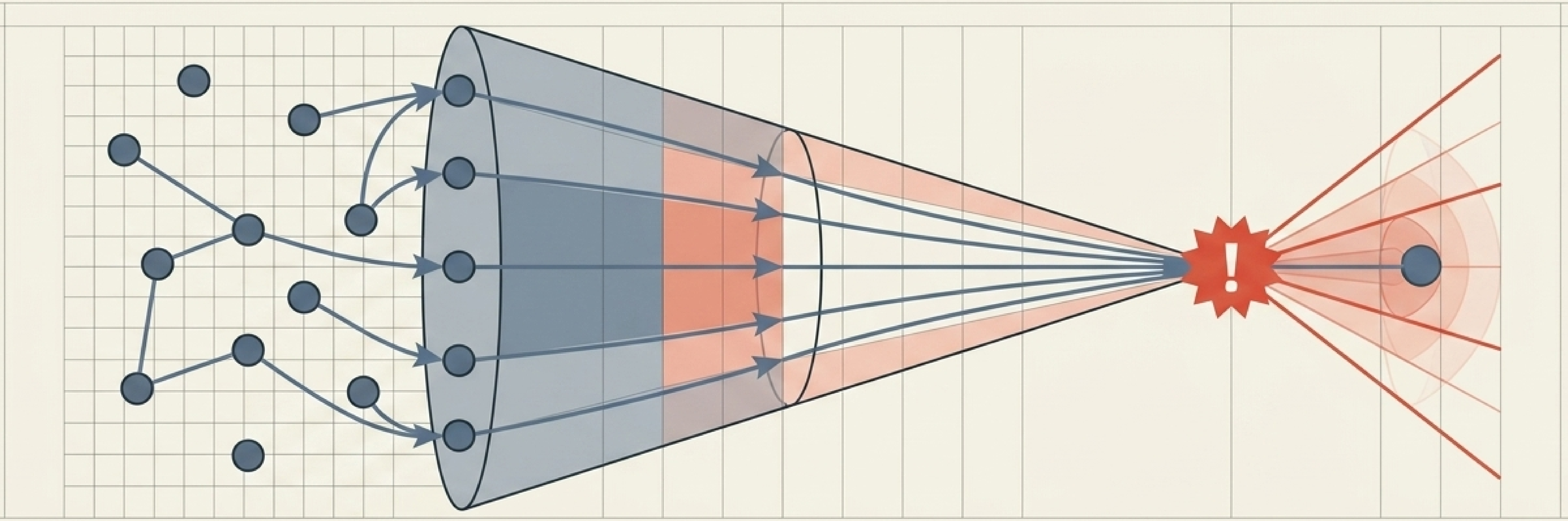
Under load, reachable futures violently diverge.
$R(x_1) \neq R(x_2)$.



Under load, reachable futures violently diverge.
$R(x_1) \neq R(x_2)$.

Reachability is a global geometric property, not a local observational one.
Hidden structural variables determine the future without affecting current observation distributions.

Corollary 1: The Predictive Gap



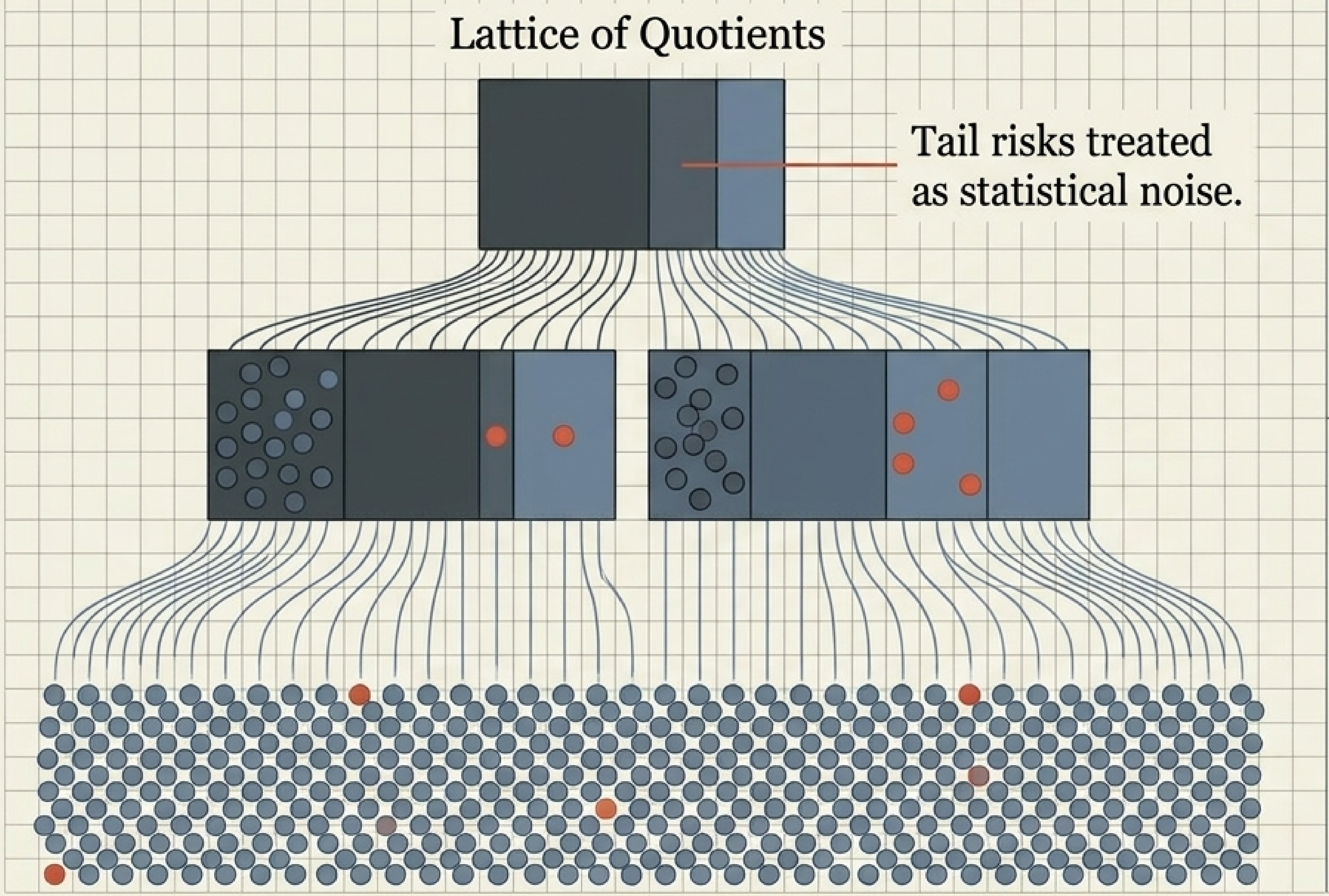
The Mechanism

Predictive representations minimize predictive loss. Therefore, they structurally force the collapse of observationally equivalent states.

The Consequence

By Theorem 4, this guarantees admissibility distortion. Predictive sufficiency does not equal Admissibility sufficiency. A model can perfectly predict the next latent state and still catastrophically fail at planning.

Corollary 2: The Compression Gap



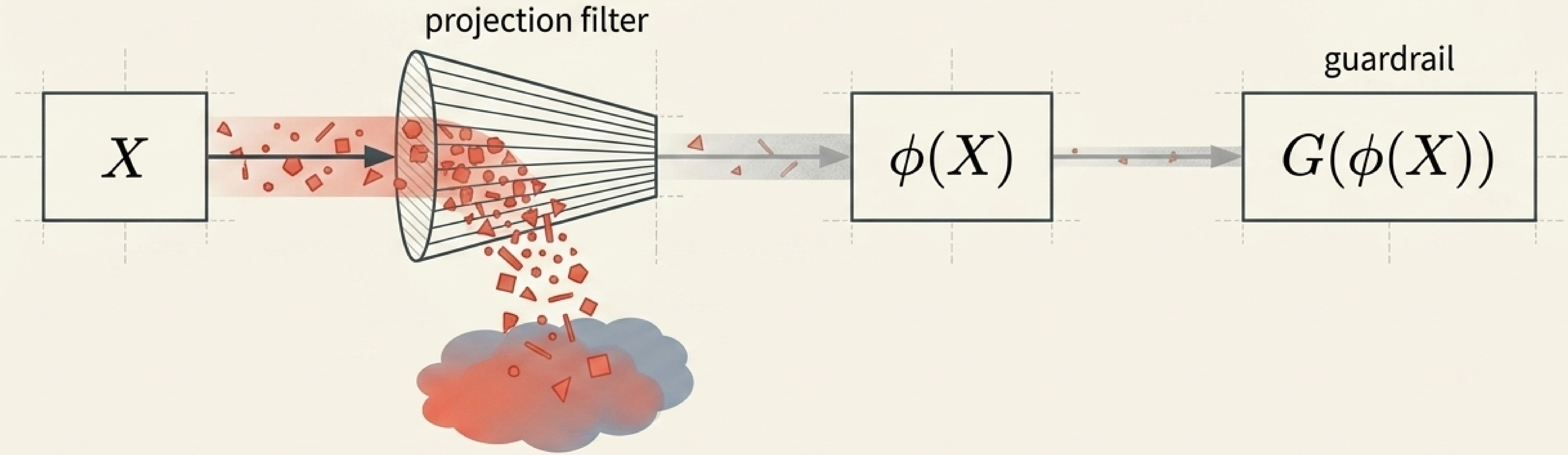
The Mechanism

Compression rewards enlarging equivalence classes. It minimizes description length by treating rare, high-consequence states as noise to be eliminated.

The Consequence

Compression exerts active mathematical pressure toward admissibility-violating collapses. Moving upward in compression guarantees walking toward higher Admissibility Distortion.

Corollary 3: Irrecoverable Quotient Collapse



The Data-Processing Inequality

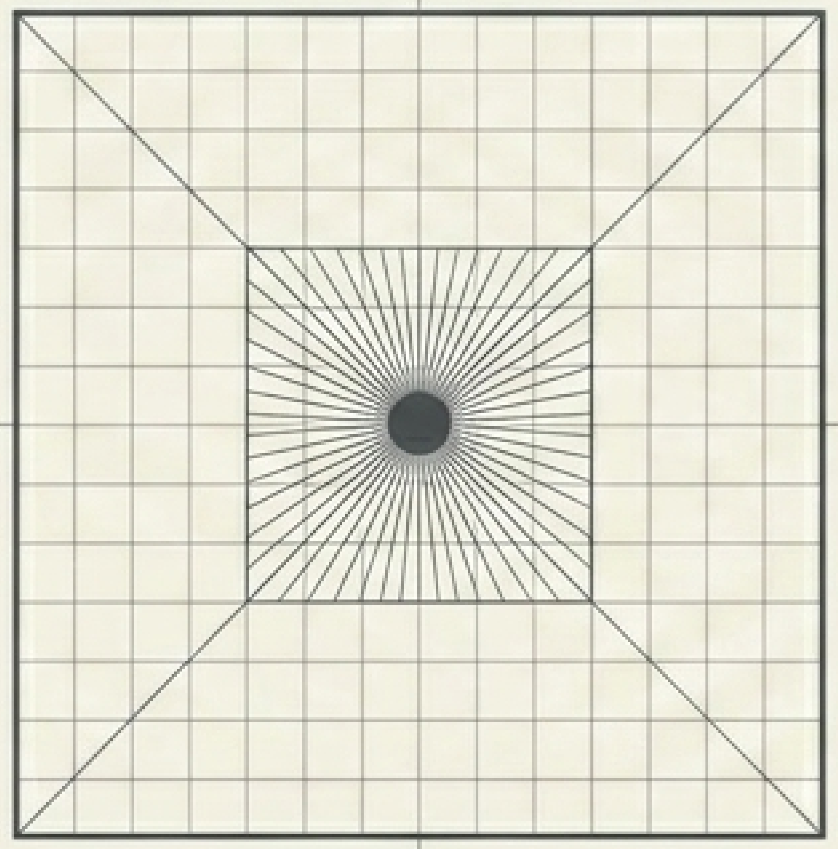
$$I(A; G(\phi(X))) \leq I(A; \phi(X))$$

The Consequence

If safety-relevant geometry is destroyed by the initial latent projection, no downstream safety head or guardrail classifier can ever recover it. The failure precedes the optimizer.

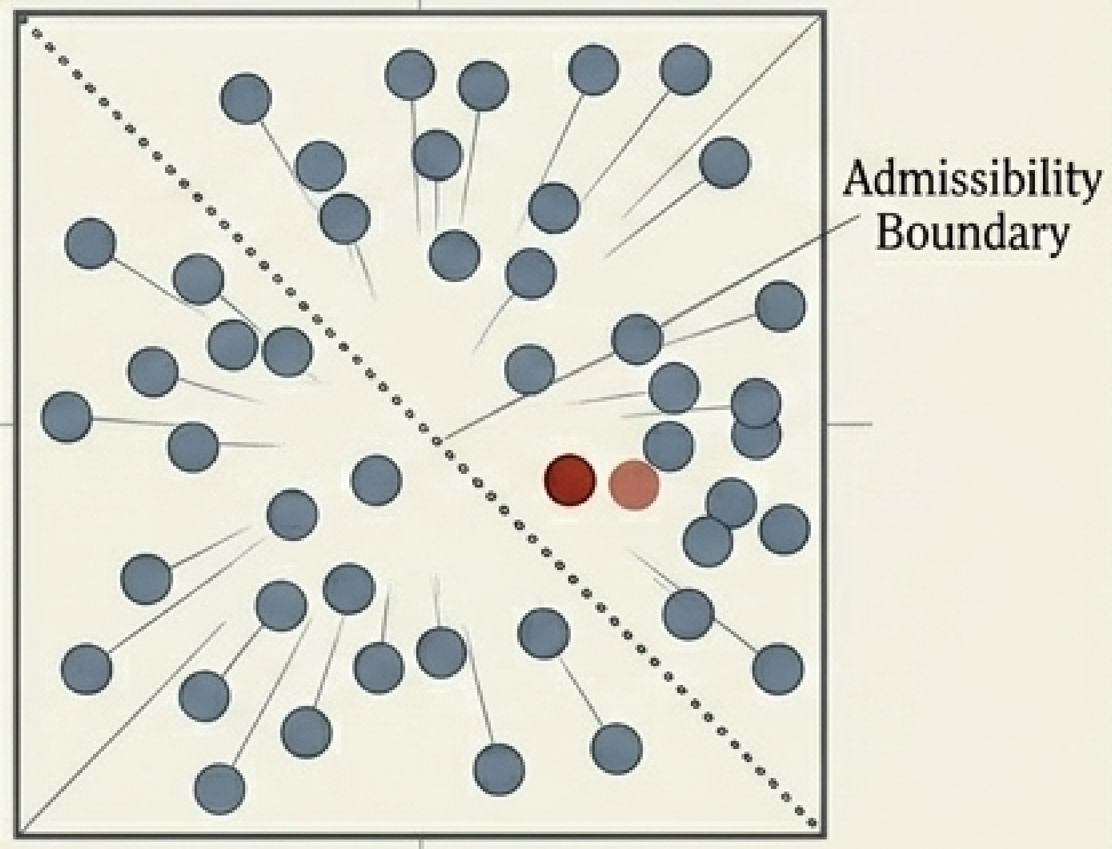
The Illusion of Anti-Collapse

Trivial Collapse (Solved)



Current methods (contrastive terms, variance regularizers) successfully prevent all states from mapping to a single point.

Geometric Collapse (Unsolved)



Spreading representations globally does nothing to prevent local, fatal misidentifications of reachability. Current methods ignore boundary crossings entirely.

Synthesis: Representation Learning as Quotient Selection

$\mathcal{X}/\sim_O$

Predictive Methods (JEPAs)

Selects quotient focusing on matching passive distributions.

$\mathcal{X}/\sim_C$

Compressive Methods (Bottlenecks)

Selects quotient focusing on minimum description length.

$\mathcal{X}/\sim_A$

Admissibility (Planning Requirement)

Requires quotient focusing on reachability geometry.

The choice of a representation method is not merely a choice of neural architecture. It is a choice of an equivalence relation over reality.

Characterization: When is Prediction Safe?

Proposition 4

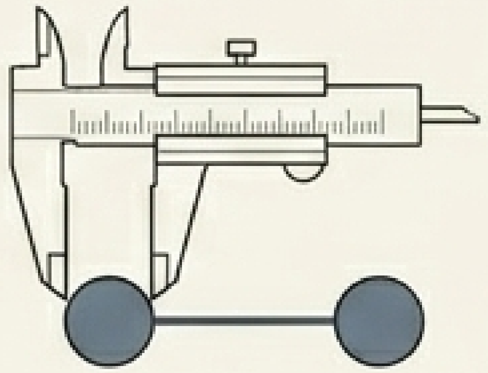
Prediction is safe IF AND ONLY IF observational equivalence is a strict subset of admissibility equivalence.

The Real-World Meaning

This only occurs when every single intervention-relevant variable is perfectly exposed in the passive observation distribution. If there are any hidden structural states, states, dormant failure modes, or unobservable causal parents, prediction will fail at planning.

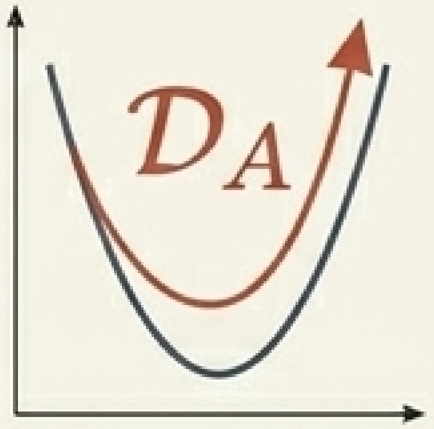
The Research Agenda: Regularizing Admissibility

Estimation



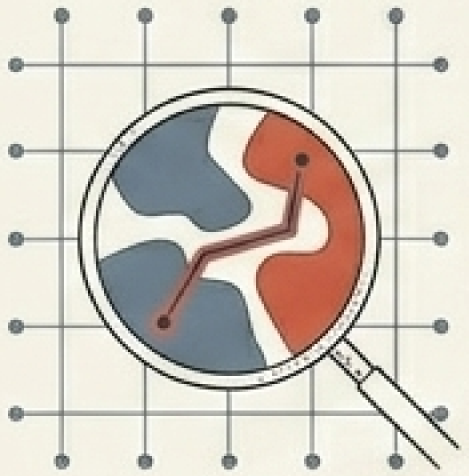
Using Bisimulation metrics as computable lower bounds for reachability divergence.

Regularization



Explicitly adding Admissibility Distortion to the training loss to penalize unsafe collapses.

Auditing



Post-hoc stress-testing of trained encoders against reachability probes in high-stakes regions.

Shift from asking “does this model predict well?” to “does this model’s latent projection factor through the admissibility quotient?”

The Missing Criterion, Found

Planning induces Admissibility ($\sim_A$)

Admissibility induces Distortion ($\mathcal{D}_A$)

Prediction and Compression are merely flawed approximations of ($\mathcal{D}_A$).

Admissibility Distortion names the exact quantity that world models currently fail to preserve. It provides the formal, mathematical criterion for evaluating if an AI truly understands the geometry of its environment—moving beyond "what is this?" to "what can I do from here?"